

Correlating biogeography and cladogenesis: How Earth's geographical transformation from Pangea to today's continents is related to 260 million years of turtle evolution

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Introduction

Can changes in Earth's geography over the last 260 million years be correlated with cladogenesis events within the clade Testudines?

This study examines the biogeographical evolution of the clade Testudines, which contains over 200 extant turtle species that are distributed all over the planet. Biogeography is considered a heritable character, where character states (CS) are geographical distribution. Like other heritable characters, biogeographical character states may be traced throughout the course of evolutionary history while studying cladogenesis. Extinct common ancestors to the testudines clade are dated back to approximately 260 million years ago (MYA)¹, the same time that the Earth's continents formed Pangea². The modern extant clade Testudines arose during the Jurassic era 125 MYA, a time period where the continents were mostly split apart, but did not yet resemble today's geography². Therefore, we suspect that some cladogenesis events may reflect geological events such as shifting land masses or mountain range formation.

Our primary hypothesis states that taxa sharing a more recent common ancestor (MRCA) will exhibit more similar biogeographical character states relative to more distantly related taxa. We also hypothesized that cladogenesis events along turtle evolution may be correlated to major geographic changes on the earth's surface. Our study is the first to simultaneously use a wide range of terrestrial and marine extant & extinct turtle species, examine nesting location as a biogeographical CS, and examine both biogeography and fossil calibrations to correlate cladogenesis with changes in Earth's geography.

Methods

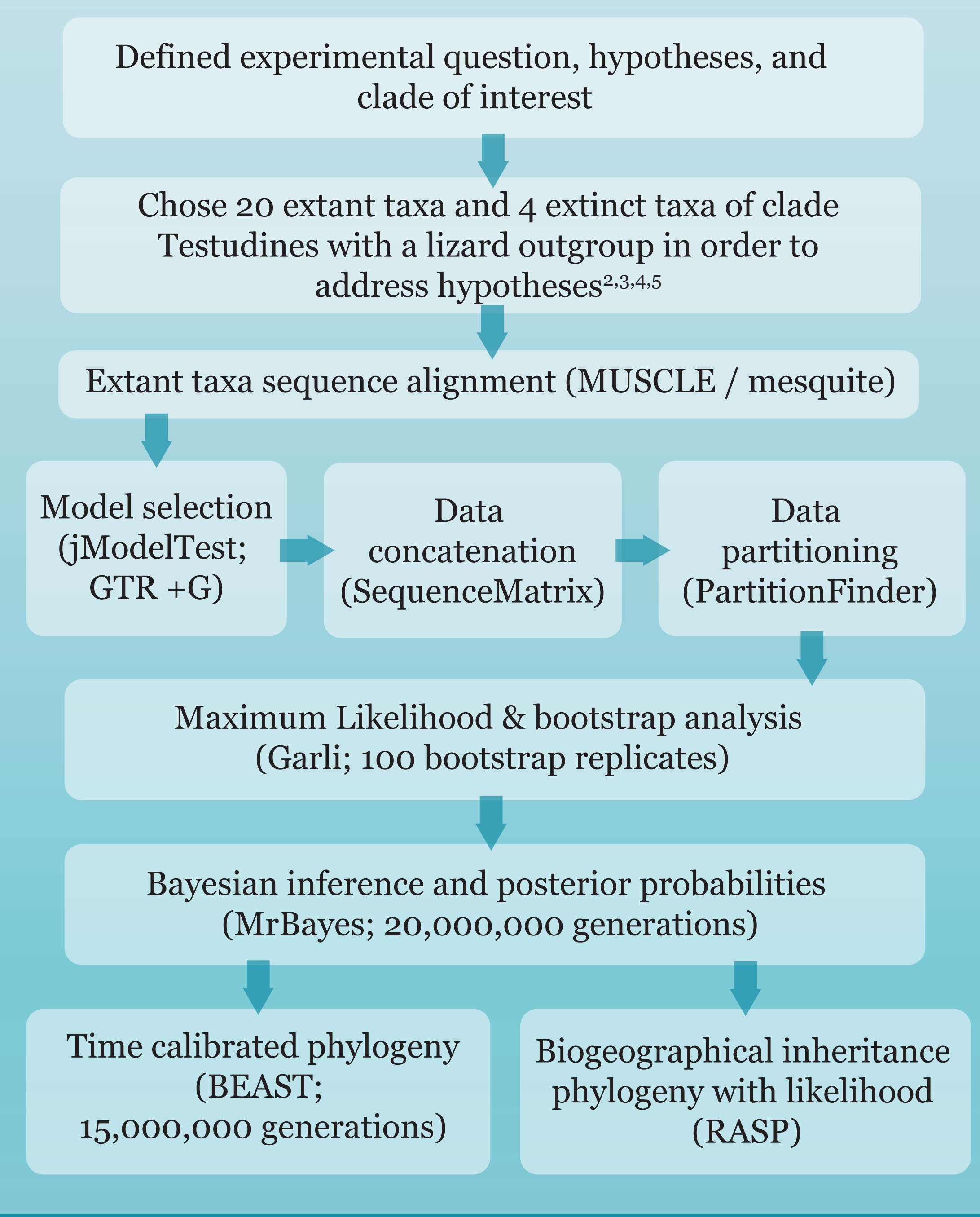


Figure 1. Time-calibrated phylogeny (chronogram) based on molecular dating estimates depicting evolutionary relationships between 20 extant members of the clade testudines and 4 extinct species. Tree is rooted with a lizard outgroup. Solid branches produced using BEAST software; dashed branches were added by hand using published fossil data.

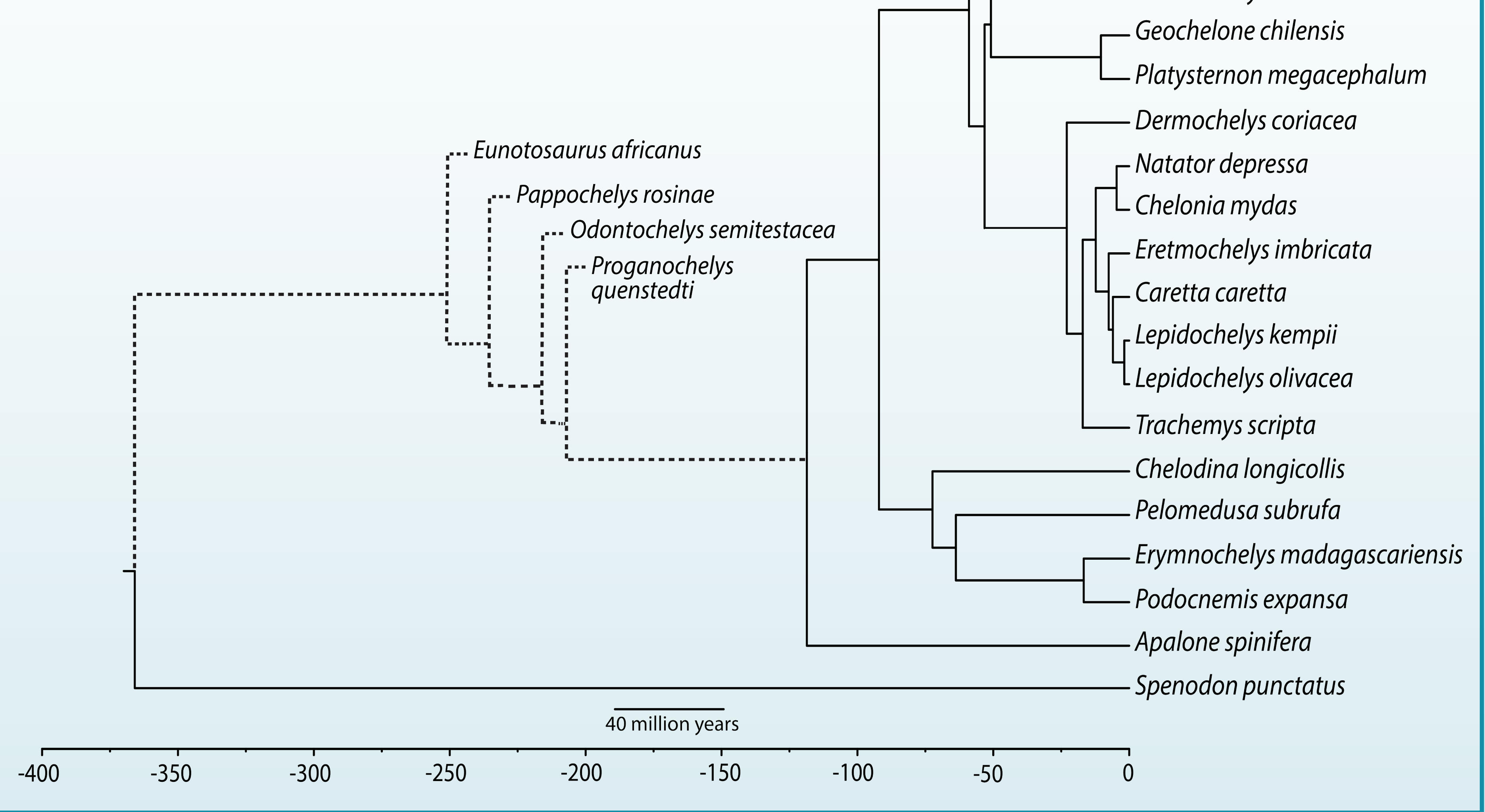
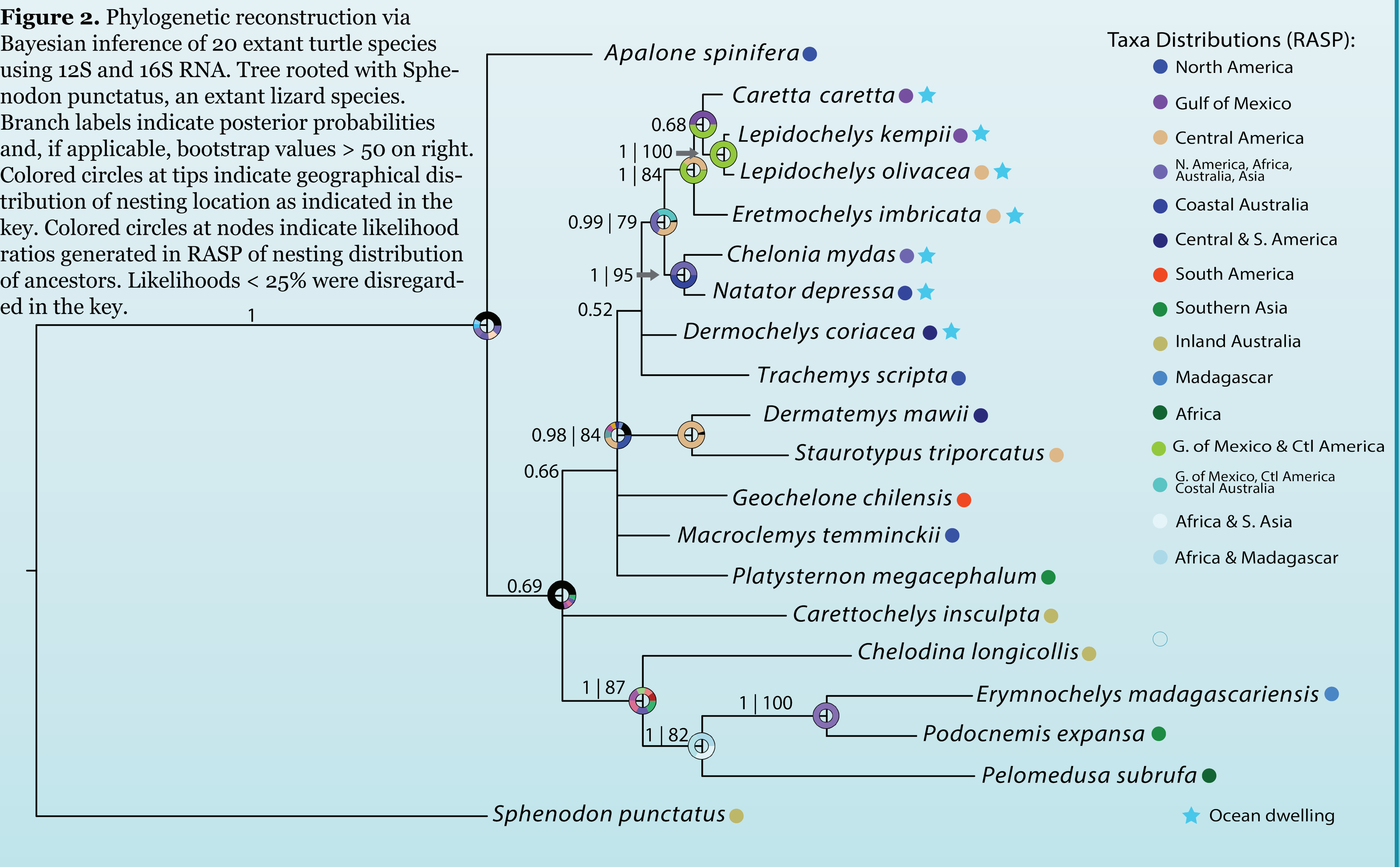


Figure 2. Phylogenetic reconstruction via Bayesian inference of 20 extant turtle species using 12S and 16S RNA. Tree rooted with *Sphenodon punctatus*, an extant lizard species. Branch labels indicate posterior probabilities and, if applicable, bootstrap values > 50 on right. Colored circles at tips indicate geographical distribution of nesting location as indicated in the key. Colored circles at nodes indicate likelihood ratios generated in RASP of nesting distribution of ancestors. Likelihoods < 25% were disregarded in the key.



Literature Cited

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Results

- No conflicts occurred between gene trees regarding the methods of reconstruction that we employed. There is correlation between relatedness of taxa and similarity of biogeographical character states (Fig. 2).
- Speciation events appear to occur at times when now-distant geographical locations were significantly closer. For ex., consider the cladogenesis event concerning *E. madagascariensis*, *P. expansa*, and *P. subrufa*, who nest in Madagascar, S. Asia, and Africa, respectively. According to Fig. 1, the event that made *P. subrufa* sister to *E. madagascariensis* and *P. expansa* occurred approximately 70 MYA. At this time, Madagascar, Africa, and S. Asia were considerably closer in proximity than today. We can attribute this turtle evolution to the major geographic changes that occurred regarding the separation of these land masses.
- Our topology involves marine turtles that share common ancestors but have vastly different range sizes. For ex., *C. Mydas* and *N. Depressa* share a MRCA that arose 5-10 MYA. *C. Mydas* is widely distributed across the Gulf of Mexico, Pacific Central America, S. Pacific Asian islands, and N. coastal Australia. Interestingly, *N. Depressa* is only located on the coasts of N. Australia. There was likely a cladogenesis event in coastal Australia that led to a speciation event where *N. Depressa* has an extremely limited distribution.
- *L. olivacea* is widely distributed across the globe and nests on almost every beach. *L. kempii*, however, is only located in N. American Atlantic waters and nests only within the Gulf of Mexico. A cladogenesis event likely happened in the Gulf of Mexico, ultimately resulting in this narrow distribution of *L. kempii*.

Discussion

- While we did not obtain support values higher than values previously reported, the topologies of the phylogenies reported in all referenced studies are consistent with ours^{4,6,7,8}.
- We began this study with the question: how is biogeography represented in 20 extant and 7 extinct turtle species? As we progressed through research, we narrowed the focal question of our project: is biogeographical distribution a heritable trait for members of testudines? We hypothesized that taxa who share a MRCA would exhibit more similar biogeographical character states relative to more distantly related taxa.
- We found that there exist patterns of cladogenesis and geographical inheritance that correlate with historical events of geological change. Our research builds on previous studies^{4,6,7,8} by providing a historical biogeographical context that attempts to explain speciation events that would seem impossible if the current state of continental separation was the case at these times of speciation. Our study also builds on the findings of Hirayama *et al.* 2000 by providing a marine species dimension to the study of turtle biogeography, as well as consideration of common ancestry of included taxa⁹.
- Future studies could focus on a specific species or examine a dispersal or vicariance hypothesis. For example, why are some taxa only located within the Gulf of Mexico, while others both live and reproduce on both the Pacific and Atlantic sides of Central and S. America? Why do some taxa live on both sides of S. and Central America, but only reproduce within the Gulf of Mexico? It may also be interesting to look at biogeography and speciation surrounding Australia, S. Asia, and SE Asian Pacific islands, as the distribution and breeding locations in this area varies greatly among sea turtle species.